Port Command: Pitch Document for MMC Ports

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The pitch in one paragraph

Port Command (working name, change it freely) is an AI operations layer that shows a whole port as one live map and proposes the decisions a control room makes by hand today.

It is built for MMC Ports: five sea ports plus SPT Services and Andaman Port, run as one network. We do not replace a terminal operating system or crane automation. MMC is already building that foundation with NIZE, the technology arm of Shanghai International Port Group, so we sit on top of it as the decision layer.

The stage line: **one live map of every berth, yard and worker, and an AI that tells the control room what to do next.**

Three promises, each visible in the demo:

- **See.** A 2D circuit-board map of the port with vessels, vehicles, containers and worker pins moving in real time.
- **Decide.** An AI copilot that explains what is happening, predicts delays and proposes berth, yard and equipment changes for a human to approve.
- **Prove.** A safety and compliance log mapped to Malaysian rules, so an audit is a query instead of a paper chase.

The prototype runs on a simulated port. Say that plainly on stage: it is a concept demo of the decision layer, not a live deployment.

MMC Ports in one page

MMC Ports is Malaysia's largest port operator: five sea ports, more than 20 million TEU handled in 2025, wholly owned by MMC Corporation Berhad (MMC Ports 2025 results).

Port	Note	2025 figure
Port of Tanjung Pelepas (PTP), Johor	Joint venture of MMC Ports and APM Terminals; transshipment hub	14,028,375 TEU, first Malaysian terminal above 14 million (MMC Ports news)

Port	Note	2025 figure
Northport, Port Klang	Container and conventional cargo	Record 3.80 million TEU; 12.90 million FWT conventional
Johor Port, Johor	Containers and liquid cargo; two new liquid jetties commissioned	Record 1.08 million TEU
Penang Port	Containers and the Penang ferry service	3.13 million ferry passengers, up from 2.87 million in 2024
Tanjung Bruas Port, Melaka	Small container operation	31,032 TEU, highest since container handling began in 2019

Say "five sea ports" on stage, not six or seven. MMC's own description reaches seven by adding SPT Services and Andaman Port in Kedah, and it also runs cruise terminals at Port Klang and Langkawi (MMC Ports overview).

People. Azman Shah is the CEO. Sultan Ahmed Bin Sulayem was reported in August 2026 as Executive Chairman (NST); confirm his exact title before naming him on stage.

Three facts worth saying out loud

1. The network passed 20 million TEU in 2025, up from 18.5 million in 2024 (The Edge).
2. PTP was the first Malaysian terminal past 14 million TEU.
3. MMC's CEO has named AI in yard planning and equipment deployment as the game-changer the group is working toward (Bernama).

Why now: MMC is already moving, and we fit its plan

MMC Ports has started a five-year digital programme, so the pitch is to plug into it, not to compete with it.

Date	Event	What it means for us
8 Oct 2026	PTP completes its electric prime mover fleet; the first 21 entered service in August (Bernama)	The energy panel has real equipment to show
8 Jul 2026	Transport Minister says the Malaysia Maritime Single Window moves to Phase Two: cargo, port services, approvals, e-payment (NST)	Vessel clearance status is a natural data feed

Date	Event	What it means for us
24 Jun 2026	PTP signs an MoU with Siemens for pilots and operational technology integration trials (MMC Ports)	MMC is open to structured pilots
21 Jan 2026	MMC Ports calls 2026 a consolidation year under its Digital Blueprint Roadmap (Bernama)	Our copilot targets the AI yard-planning goal the CEO named
18 Dec 2025	MoU with NIZE: five-year roadmap, group dashboard with drill-down to each terminal, digital twin pilot at one port (The Star)	Our group portal and map are the front end such a twin needs

Two national items are still ahead: a Malaysia Maritime Masterplan 2026 to 2040, a 15-year roadmap on decarbonisation, digitalisation and fleet, and a revision of federal port law that the Transport Minister has targeted for the end of 2026 (Bernama).

The pitch line: MMC has chosen the foundation; we show what an AI decision layer does on top of it.

The problem and our answer

A control room still stitches vessel schedules, yard plans, equipment status, gate traffic and safety reports together from separate screens, radios and spreadsheets; Port Command puts them on one map and adds an AI that acts on them.

Where manual work typically remains (check each point with someone who has worked a terminal):

- Berth and yard plans re-done by phone when a vessel runs late.
- Equipment dispatch: which crane, truck or prime mover goes where.
- Container position checks and tally reconciliation.
- Safety permits, incident reports and contractor sign-ins.
- Compliance reports assembled after the fact.

A port network is too large for one person to watch, so Port Command is a set of role-based portals over one shared state:

Portal	Main user	Job
Terminal Control (the circuit-board map)	Duty manager	One live picture of vessels, vehicles, containers and workers

Portal	Main user	Job
Vessel and Berth Planning	Berth planner	Arrivals, ETAs, berth windows, delay predictions
Yard and Equipment	Yard planner, equipment controller	Container positions, crane and truck utilisation, dispatch proposals
Safety and Workforce	HSE officer	Zones, live alerts, contractor tracking, incident log
Gate and Landside	Gate and haulier teams	Truck queues, appointments, gate throughput
Compliance and Audit	Compliance lead	Rule-by-rule evidence, exportable audit trail
Energy and Sustainability	Operations head	Electric equipment, charging, emissions per vessel call
Cyber and OT Security	Security team	Alerts, asset status, incident notification clock
Shipping Line Portal	Lines and agents	Their vessels, container status, berth slot
Group Dashboard	MMC Ports executives	All ports side by side, drill-down to any terminal

The AI copilot works across all of them at three levels of autonomy, and the demo should show each: **inform** (explain what is happening), **recommend** (propose a change with its reasoning) and **act with approval** (a human confirms before anything executes). The human approval step is what makes the safety and regulatory story credible.

Screens, widgets and demo behaviour for each portal are in the prototype specification.

The numbers, and how to defend 100 to 10

The best documented result is Qingdao's: about 60 workers per vessel down to 9, an 85% cut, and it came with automated cranes and vehicles, not software alone.

Benchmark	Result	Source
Qingdao New Qianwan terminal, opened 11 May 2017	About 60 workers per vessel down to 9; labour cost down 70%; efficiency up 30% because it can run at night	Maritime Executive
Tianjin smart terminal	Built in 33 months; single-bridge efficiency up over 40%; personnel allocation down 60% against a traditional terminal	Hellenic Shipping News
Shandong Port upgrades of existing terminals	Overall cost 70% lower than building a new automated terminal	CGTN via AAP
Industry estimate (2018)	A fully automated terminal needs at least 45% fewer workers	iContainers

The Shandong row matters most for MMC: its ports already operate, so the relevant story is upgrading a working terminal, not building a new one.

How to say 100 to 10 without being caught out

- Call it an illustrative target for the roles a control layer touches, and label the slide "Illustrative model, to be validated with MMC data".
- Anchor it to Qingdao (60 to 9) and give a range: 45% to 60% as the conservative band, up to about 85% as the best documented case.
- Show the mechanism, not only the ratio: which coordination, monitoring and paperwork roles shrink, and why.
- Say outright that the largest reductions in the benchmarks came with automated hardware, which is MMC's capital decision. Our layer raises utilisation and cuts coordination work on whatever equipment mix a port has.

The benchmarks date from 2017 to 2022 and are mostly Chinese terminals. They show what is possible, not what MMC would get.

Regulation hooks

No single new Malaysian rule mandates a system like this; the honest pitch is that recent rules raise the cost of manual compliance and Port Command lowers it. This is not legal advice, so have counsel check any legal reading before it goes on a slide.

Rule	What it requires (verified)	Feature it maps to	Pitch line
Occupational Safety and Health (Amendment) Act 2022, in force 1 Jun 2024	Covers all workplaces; employers must ensure the safety of contractors and subcontractors; new OSH coordinator role; certificate-of-fitness rules for certain plant (Rajah and Tann)	Live exclusion-zone alerts, contractor tracking, equipment certificate expiry tracker	"Every person on the quay, employee or contractor, sits inside one safety picture."
Cyber Security Act 2024, in force 26 Aug 2024	Transportation is one of 11 critical sectors; designated entities follow codes of practice, run risk assessments and audits, and report incidents (NST, EY)	Cyber and OT Security portal with incident log and audit trail	"If a port is designated critical infrastructure, the audit trail is already running."
Personal Data Protection (Amendment) Act 2024, phased 1 Jan to 1 Jun 2025	Biometric data becomes sensitive data; mandatory breach notification; data protection officer; direct duties on processors (Baker McKenzie, Conventus Law)	Worker pins limited to shift hours, zone-level view for most roles, access log, privacy officer view	"We track for safety, not surveillance."
Malaysia Maritime Single Window, Phase Two	Extends from vessel clearance to cargo processes, port services, approvals and e-payment (NST)	Vessel clearance status on the map; an integration point	"Our map speaks the language of the national single window."
Maritime Masterplan 2026 to 2040 and port law reform	Decarbonisation and digitalisation anchor the 15-year roadmap; federal port law revision targeted for end 2026 (Bernama)	Energy portal: electric equipment, emissions per vessel call	"Built for where the regulation is heading."

Handle the worker pins with care: location tracking of named people is personal data, so show the privacy controls in the demo, not only in the slides.

Not researched here: ISPS Code port security rules and IMO requirements. Each could add one row if time allows.

Stage script (five minutes)

Spend about three of the five minutes inside the live map and let the numbers come last. The timings assume the scripted scenes in the prototype specification.

Time	Beat	What to say	What to show
0:00 to 0:30	Hook	"In 2017 Qingdao unloaded a ship with 9 workers where it used to need about 60. MMC Ports moved over 20 million containers last year across five ports. Who watches all of it?"	Port map idle, vessels slowly moving
0:30 to 1:00	Problem	"Today planners juggle separate screens, radios and spreadsheets."	One slide, or the map with the portal tabs dimmed
1:00 to 1:45	See	"One live map of the terminal. Every blue dot is a person, every block a vehicle. This is simulated data."	Zoom in and out; click a vessel (origin, destination, ETA); click a worker pin (task, zone, shift)
1:45 to 2:30	Docking	"A vessel is entering the docking zone. The system has already assigned cranes and routed vehicles."	Ship slides into the berth; crane and truck animation
2:30 to 3:15	Safety	"A worker has entered a crane exclusion zone. The system holds the crane and logs it for the safety audit."	Red alert, one-click approval, log entry
3:15 to 4:00	Decide	Ask the copilot: "Why is Berth 2 behind?" then "What should we change?"	Answer with reasoning and a proposal; click approve
4:00 to 4:30	Scale	"MMC runs five sea ports. One group view, with drill-down to any terminal."	Group Dashboard

Time	Beat	What to say	What to show
4:30 to 5:00	Impact and ask	"Qingdao shows up to 85% fewer people per vessel call. We model 100 to 10 as an illustrative target for control roles. We would like to pilot on one berth."	Illustrative model slide, then contact slide

Say "simulated" once, early. Judges forgive a simulation; they do not forgive a surprise.

Record the full demo as a screen video before the event. If the laptop or network fails on stage, the video keeps the five minutes intact.

Likely questions and honest answers

Every answer below stays inside what the research supports; where something is a design choice rather than a fact, it says so.

Question	Short answer
Isn't this what NIZE and Shanghai already do for MMC?	NIZE brings the architecture, the group dashboard and a digital twin pilot. Port Command is the AI decision layer on top: it explains, predicts and proposes. We would plug into that foundation, not replace it.
Is the data real?	No, the prototype runs a simulated port. In production it would read vessel positions from AIS, terminal data from the operating system, equipment telematics, worker badges and the national single window.
Where does 100 to 10 come from?	It is an illustrative target. The best documented case is Qingdao, about 60 to 9 per vessel, achieved with automated hardware. Our layer cuts coordination and monitoring roles on any equipment mix, and a pilot would measure it.
What happens to the workers?	The target is roles, not people: redeployment into supervision, maintenance and data work, phased through attrition and retraining. Malaysian maritime workforce discussions treat people and skills as a core theme of transformation (TransNav).

Question	Short answer
Isn't tracking workers a privacy problem?	Pins run only during shift, most roles see zone-level positions, every view is logged, and a data protection officer has an audit view. That is our design, built around the 2024 PDPA amendments.
What if the AI is wrong?	It proposes with its reasoning, a human approves anything that moves equipment, and everything is logged. In phase one it never controls equipment directly.
What about cybersecurity?	Transportation is a designated critical sector under the Cyber Security Act 2024. The proposed design reads operational systems through one-way gateways first and keeps an audit trail; the Cyber portal shows that posture.
What does it cost and what is the return?	Not modelled yet, and we will not invent a figure. A pilot would track berth productivity, equipment idle time, incident response time and planner hours.
Why should MMC trust a student team?	We propose a bounded pilot on one berth, built on open-source components with no lock-in. MMC already runs structured pilots, such as the Siemens one at PTP.
How do you make money?	Not decided. The working assumption is a per-terminal subscription after a paid pilot.

Claims to avoid

The pitch stays credible if it never says the following.

- **"Nobody has done this."** Qingdao, Shanghai and Tianjin run live monitoring, and MMC has its own NIZE programme. The defensible claim is an AI decision layer for existing, mixed-equipment terminals.
- **"100 to 10" as a fact.** It is an illustrative target; the best documented case is 60 to 9, and it came with automated hardware.
- **"MMC is a designated critical infrastructure entity."** Designation under the Cyber Security Act 2024 is made by sector leads, and we found no confirmation for MMC.

- **"MMC owns six or seven ports."** Say five sea ports; seven only if you list SPT Services and Andaman Port.
- **Anything on the IPO or stake sale.** Press reports in 2026 say the listing may be scrapped and that a minority stake sale was explored; it is market-sensitive and off topic.
- **Simulated numbers as live ones.** Label every chart in the demo "simulated".
- **Real worker names or faces.** Use fictional names in the demo.

Two things to confirm before the event: the exact titles of MMC's CEO and Executive Chairman, and whether the competition expects a stated business model.

The benchmark figures are secondary reports of 2017 to 2022 results; treat them as indicative and cite the source on the slide.

Sources

Figures come from the text of these pages as retrieved on 28 October 2026; re-check the headline numbers before presenting.

MMC Ports

- MMC Ports 2025 results (repost)
- MMC Ports news and press releases
- MMC Ports overview, Breakbulk Asia
- The Edge: MMC Port history and 2024 volume
- NST: Sultan Ahmed's appointment
- The Star: NIZE MoU
- Bernama: digital consolidation in 2026
- Bernama: PTP electric prime movers and maritime policy

Benchmarks

- Maritime Executive: Qingdao
- Hellenic Shipping News: Tianjin and Qingdao
- CGTN via AAP: Shandong upgrades
- iContainers: automation estimates

Regulation and policy

- Rajah and Tann: OSH Amendment Act 2022
- NST: Cyber Security Act in force
- EY: Cyber Security Bill 2024 obligations

- Baker McKenzie: PDPA amendment dates
- Conventus Law: PDPA phases
- NST: Maritime Single Window Phase Two
- Malay Mail: Maritime Single Window rollout
- TransNav: Malaysia Maritime Week 2025 synthesis